PLA Outer Rim

Tire body V1.1

Radius of curvature of sketch1: 250mm

Arc angle between 2 center points: 5degrees (arc length: 22.51475mm)

Tire body V1.2

Radius of curvature of sketch1: 252mm

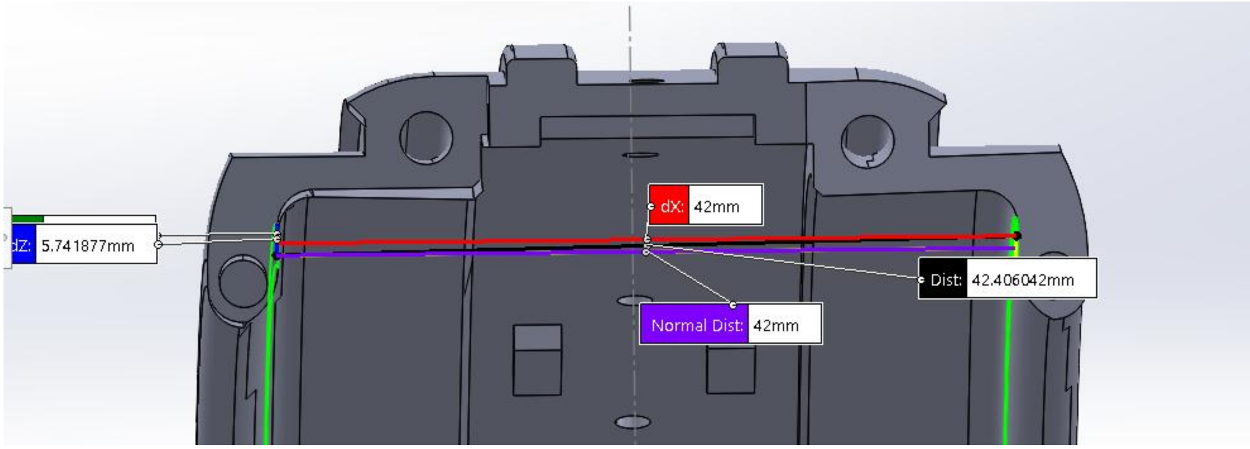
Arc angle between 2 center points: 5.3deg (arc length: 23.99mm)

Never printed v1.2, was a fix to an issue that didn’t exist.

Interference on right side

Bent

Normal



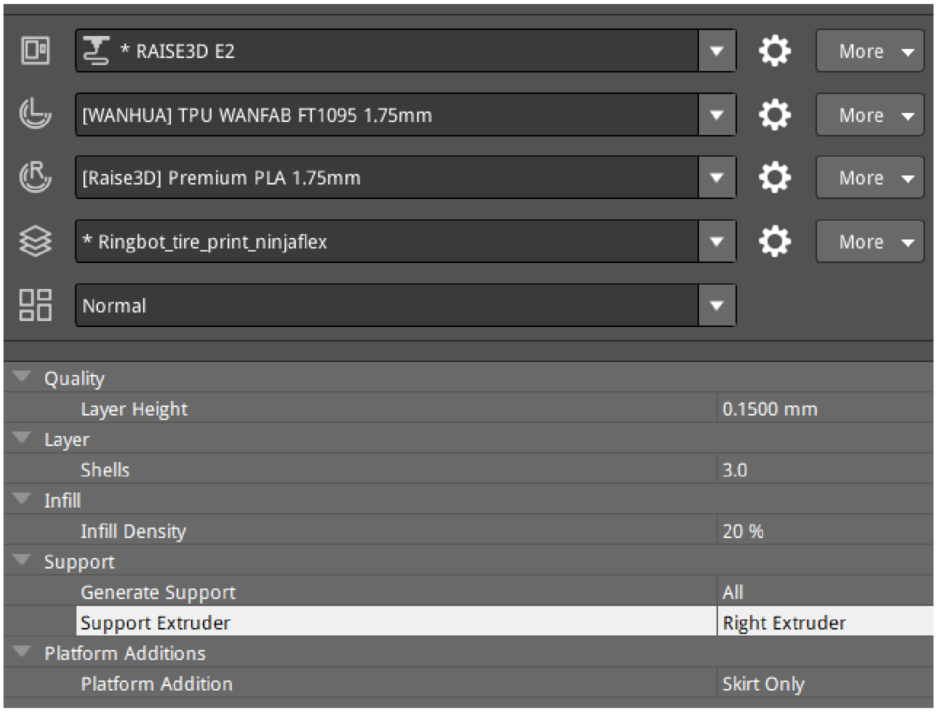
V1.1 shows 42mm, V1.1 piece measures ~41.7mm, old piece measures ~42mm. Moderate variability due to support remnants and low print quality where supports used to be connected.

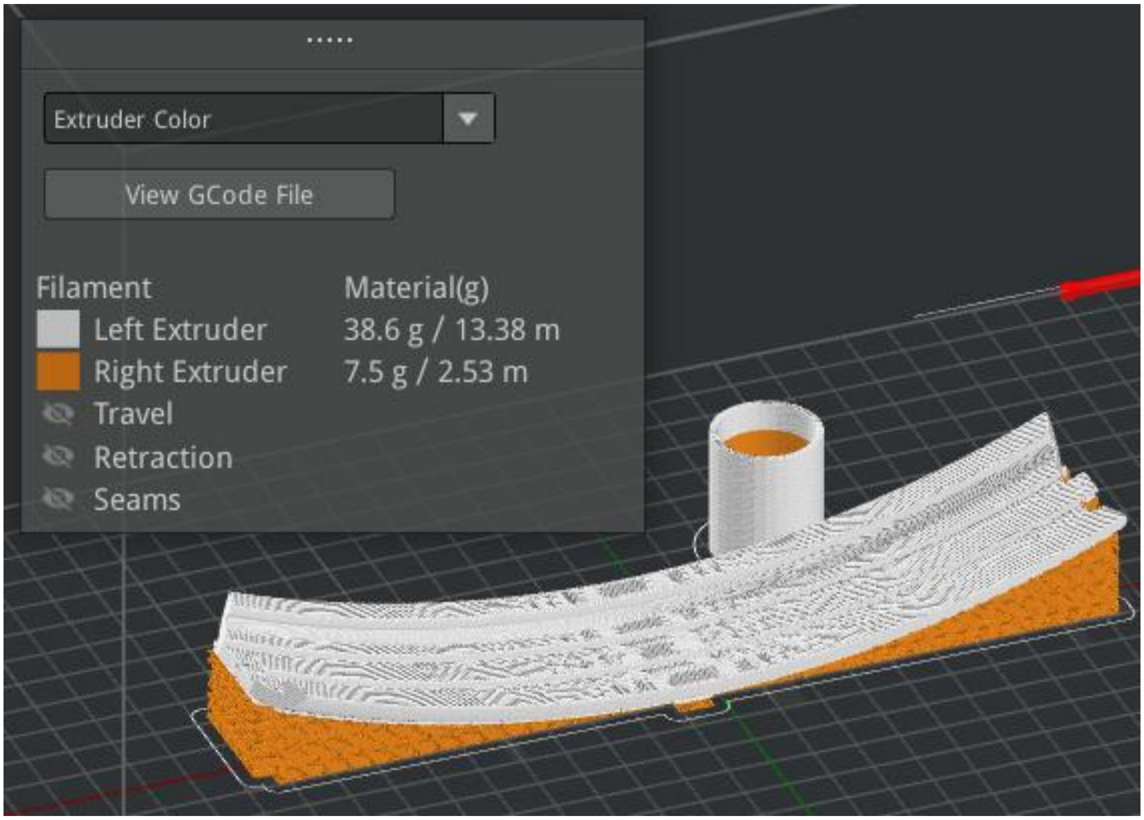
V1.3

I increased this distance to 44mm by reducing the thickness of the flange by 1mm on each side. To do this, I also had to reduce the peg diameter from 2.2mm to 1.8mm, and the hole diameter from 2.7mm to 2.2mm.

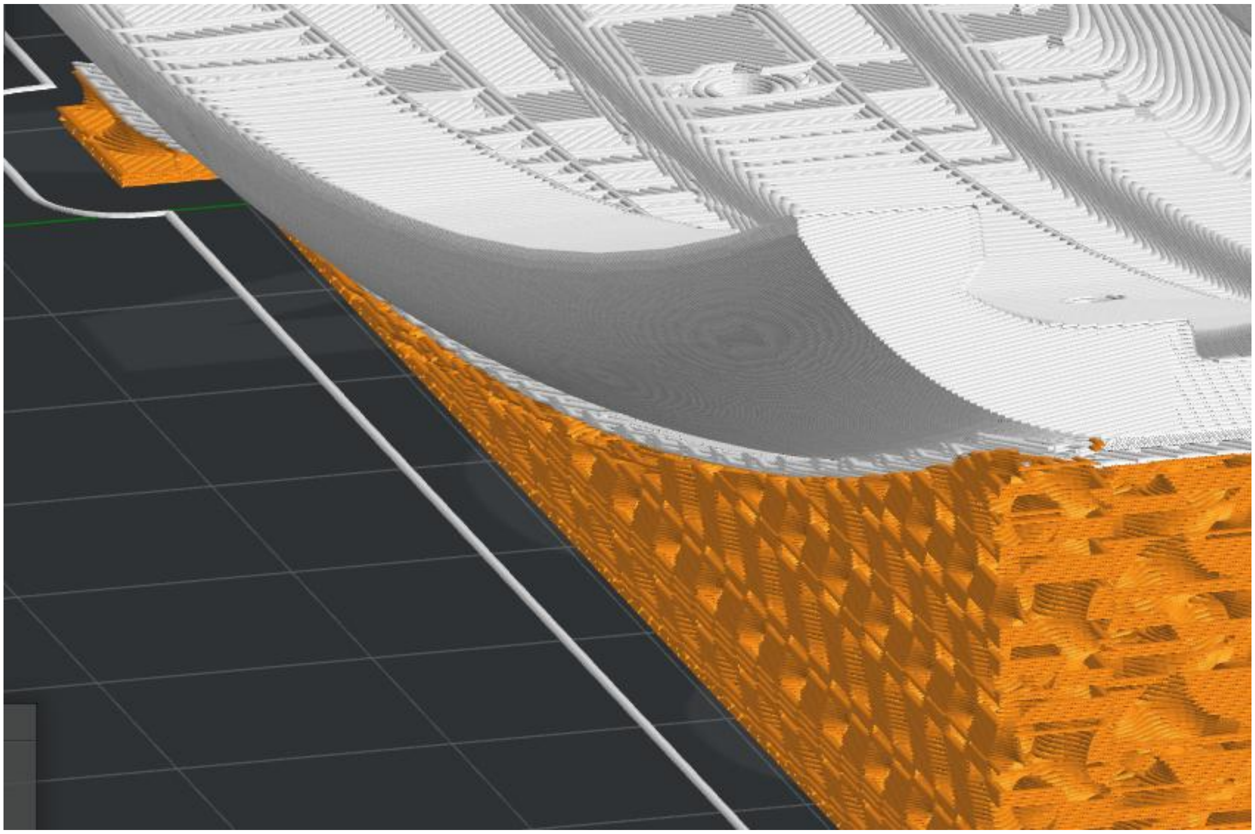
TPU Tire:

Full TPU tire print is nasty, supports are very hard to remove. Raise3D E2 printer can maybe multi extrude for PLA supports. Downloaded ideaMaker to try and print on the E2 using files and templates from Kenta.



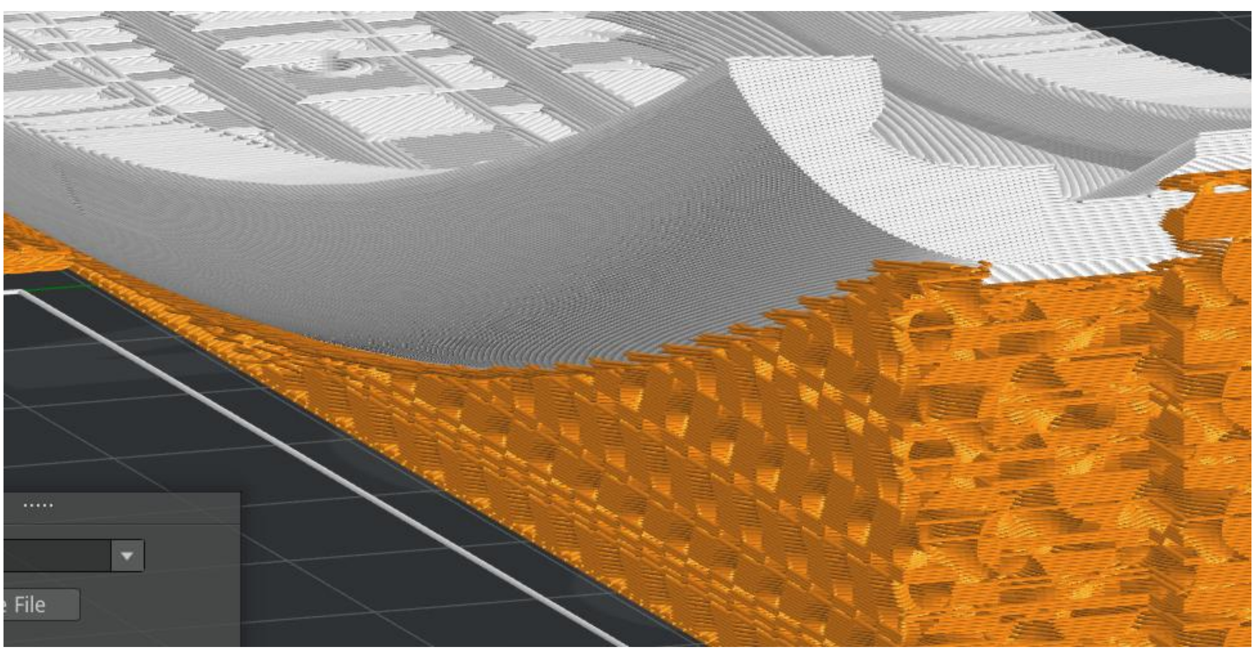


55C bed temp failed, trying 60. 60 worked, but issue was not bed temp, it was z-calibration of right extruder. TPU printed with PLA support, but there was one layer of TPU between the main body and supports.



For some reason, slicer is adding in an extra layer of TPU, as shown in white, between the main body and the PLA supports.

- Dense support material from TPU to PLA
- Vertical offset layers = 0



This change made it a bit easier to separate the supports, but it still took a decent amount of time to remove them. Either way, the new part looks nicer.

Vertical top offset layers 0 -> 1

Dense support infill 80% -> 70%

This led to easier removal, but removal time was still about 45 min – 1 hour.

Support Infill Type: Gyroid -> Lines

PLA Temperature: 225C -> 200C

Dense Support Fill: 70% -> 55%

Bits of PLA stuck in TPU

Dense Support Fill: back up to 80%

Support Infill Type: back to Gyroid

Enable Cooling Fans

This worked! PLA and TPU separated quite well. 15-20 mins per part of support removal. Printing the remaining tires with these settings.

Printing 2 pieces at once kept failing, so final 2 pieces will be printed one at a time.

Single piece failure.

Applied more glue, single piece success

Drive Module

Needed to assemble one drive module:

- 2 servos
- 2 motor shaft flange
  - Taken from motors in boxes
- 24 M1.5 screws: 4 per flange, 8 per motor for mounting
  - In little baggies in Robotis box
  - Use shorter screws for the flange, otherwise they stick out and interfere with servo
- 4 M2.5 screws: 2 for motor shaft, 2 for small gears shaft
  - M2x10 for motor shaft
  - M2x22 for small gears, couldn't find, using M2x20
- 1 left 51 tooth Herringbone
  - Printed
- 1 right 51 tooth Herringbone
  - Printed
- 1 left 14 tooth Herringbone
  - printing
- 1 right 14 tooth Herringbone
  - printing
- 2 wheel spur gear
  - printing
- 2 rings
  - Created in CAD, added chamfer to assist with bearing placement
- 2 thrust ball bearing 16mm OD 8mm ID
  - McMaster bags in Ringbot box
- 2 washers for the thrust bearings
  - Same bags as thrust bearings
- 2 radial bearing 10mm OD 3mm ID
  - Found in McMaster bags in ringbot cardboard box
- 2 radial bearing 12mm OD 8mm ID

Weird 2 ended bent Phillips head tool works for M2 and M2.5

<u>Ring housing for 10mm bearing</u>

Couldn’t find in assembly, redesigned in CAD. 10mm ID, 13mm OD, 6mm height, slight chamfer for bearing insertion

Spur Gear:

Bearing and washer nominal diameter is 16mm, current housing hole is 16.35mm, CAD dimension is 16.4mm. Changing from 16.4mm to 16mm. Printed, bearing fits, but the gear is too thick. Changing thickness from 6.4mm to 5.8mm. This worked and meshed well when mounted to the rack, at first, but then the mount became very tight. I think this is because of the thrust bearings. I’m increasing the recess hole depth by 0.3mm. This worked. Shaft hole is a bit too small for M3 screw to bite, had to use drill. Increasing from 2.6mm to 2.8mm for future use, but not going to reprint.

Drive Module Body:

Sanded support side where 14 tooth Herringbone and 25mm spur contact the body.

Large Herringbone Gear (51 teeth)

I noticed that the screws I used to attach the flange to the gear were sticking out the other end of the flange and rubbing on the servo when spinning, so I replaced them with shorter screws. This removed the screw-servo interference, but now the gear itself is rubbing on the body of the drive module. Dist from servo to gear surface is 3.9mm (calipers), while thickness of drive module body is 3.75mm (calipers). Extruding mount face of gear an extra 0.3mm. Gear and body no longer rubbing, large gear and small gear meshing remains good. Printing 2 more for second drive module.

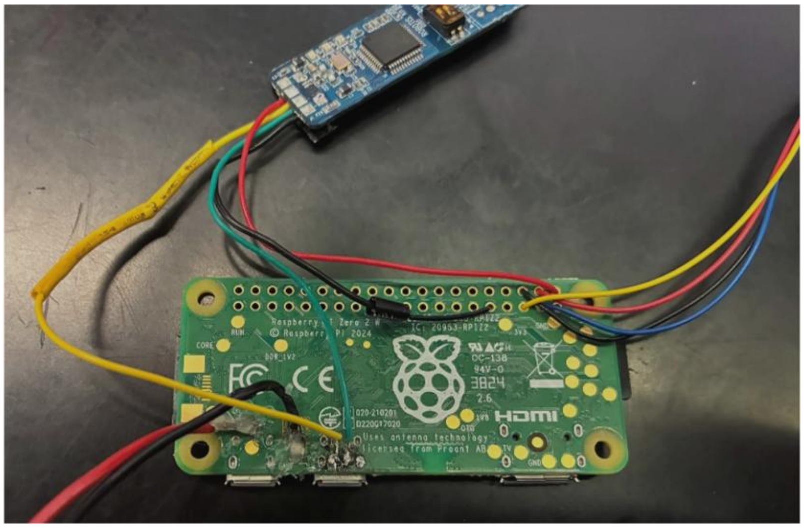
Electronics Assembly Checklist

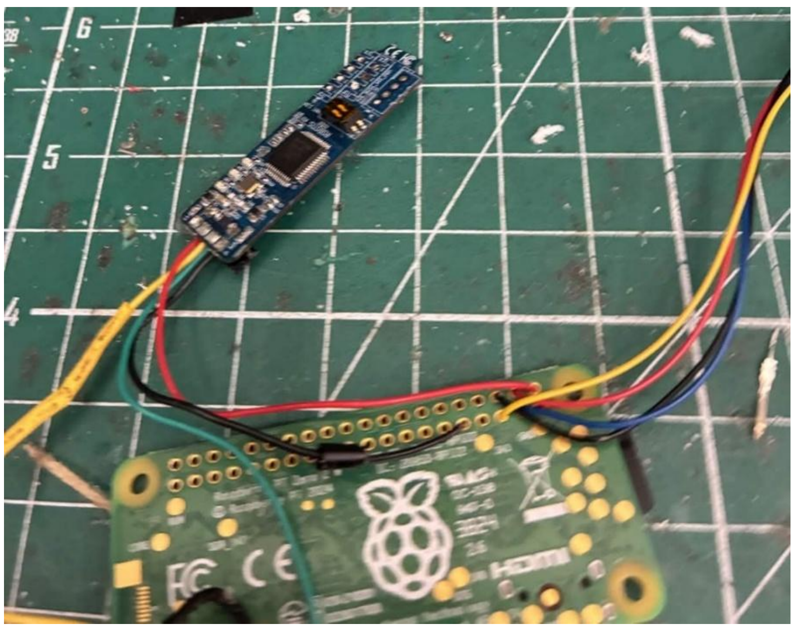
- Mechanical drive module
- BNO IMU
  - Press fit into indent between large gears
- Raspberry Pi
  - Screws in on top of IMU, SD facing towards battery
- U2D2 controller
  - Press fit into body cover
- Body cover
  - Screws into main body, big hole opposite battery side
- XM430 W210 motor
  - Bottom cover removed, rest of motors screws into body cover
- 2/1 voltage regulator standoffs
  - Screw into side of motor
- 2/1 reg19a voltage regulators
  - One module needs 2, other module needs 1
  - Screw into standoffs
- Screws
  - 2 M2.5x5mm for RPi
  - 2 M2.5x5mm for each voltage regulator standoff
  - 2 M2.5x12mm for cover

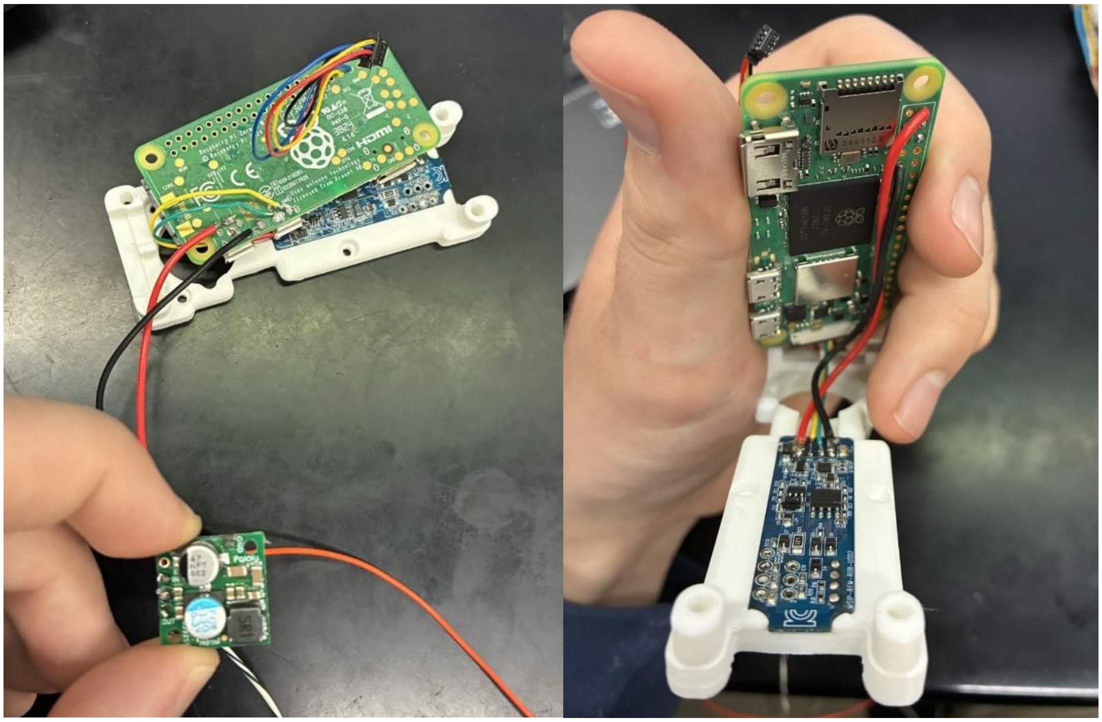
Body Cover

Divot added to account for plastic cover in leg base motor

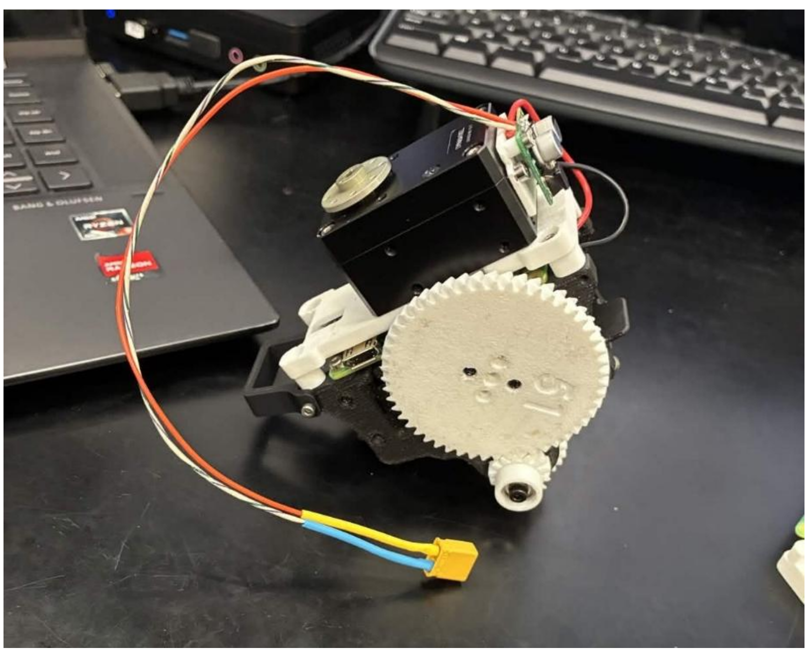
Resoldering


HDMI


6
5
4



After initial resoldering followed by many changes, tweaks, and fixes, this is the Raspberry Pi with it's connection to the voltage regulator, the U2D2 motor controller, and the IMU (not pictured, contects the 4 pin Stemma QT connector).



Partially complete drive module assembly. Still missing the headers that communicate across the telephone wire.

Simulation

1. Terminal 1:

roscore

2. Terminal 2:

source ~/ringbot_ws/devel/setup.bash

rosrun mujoco_sim ringbot_sim

3. Terminal 3:

source ~/ringbot_ws/devel/setup.bash

rosrun ring_robot_ros ringbot_control_node

To switch between sim and hardware files, swap out the files, then run

cd ~/ringbot_ws

catkin_make -Dmujoco_DIR=~/mujoco_install/lib/cmake/mujoco